HybridQC

Supplementary results and extended methods

Companion to HybridQC: Hardware-Grounded Simulation of Tightly Integrated Hybrid Quantum-Classical Systems.

This document presents the supporting analyses beyond the main paper: scheduling and topology comparisons, deployment overhead, workload growth, controller/link sensitivity, and quantum-annealing amplification. It also explains the extended model settings needed to interpret them. The numerical tables and figure data are archived in the supplementary dataset [1].

Reading the results

A hybrid computing unit (HCU) combines classical processors, quantum devices, controllers, memory, and links. Quantum annealing (QA) solves an encoded optimization problem through repeated samples. Digital quantum computing (DQC) executes gate-based circuits. A workload family is a group of jobs generated from the same application template. An experiment configuration specifies a workload, machine, policy, and scale.

The main paper defines the execution model, timing equations, workload ranges, policy rules, and principal results. Here, times are simulated seconds unless explicitly identified as hardware measurements. Completed configurations and incomplete configurations are reported separately. Quantum error correction is disabled in these studies.

Extended controller and routing choices

Controller variants cover host relay, controller-local processing, controller-to-controller links, near-device control, hierarchical coordination, direct device paths, and selected bypass modes. Each variant requires the corresponding resources and directed edges. A QA-to-DQC workflow normally returns through the controller and host before DQC submission. Shorter routes exist only when declared in the topology.

Link profiles specify latency, bandwidth, and payload movement. Available families include host accelerator links, controller/device links, and cross-modality links. Replication adds resources and their configured connections. It does not shorten the service time of an individual operation.

Optional latency variation

The jitter study varies host-link latency while preserving bandwidth. Let ℓ denote nominal latency, $J \sim \mathcal{N}(1, \mathrm{CV}^2)$ the random latency factor, and $\varepsilon > 0$ a positive latency floor. The perturbed latency and the transfer time for B bytes are

$$\ell' = \max(\varepsilon, \ell J), \tag{S1}$$

$$T_{\mathrm{transfer}}(B) = \ell' + \frac{B}{\beta}, \tag{S2}$$

where β is bandwidth in bytes/s. $\mathrm{CV} = 0.05$ gives 5% relative latency variation before the floor is applied. Main-paper link profiles are deterministic.

Data: `system_configuration/controller_variants.csv`, `system_configuration/link_profiles.csv`. All paths are relative to the supplementary dataset [1].

Data archive

[1] P. Christou, S. Kan, and Y. Mao. *HybridQC: Supplementary Data*, version 1.0.0, 2026. doi:10.5281/zenodo.23170503.

1. Scheduling and resource upgrades

Policy time composition complements the main paper's throughput and p95 frontier by showing the classical, controller, and quantum components.

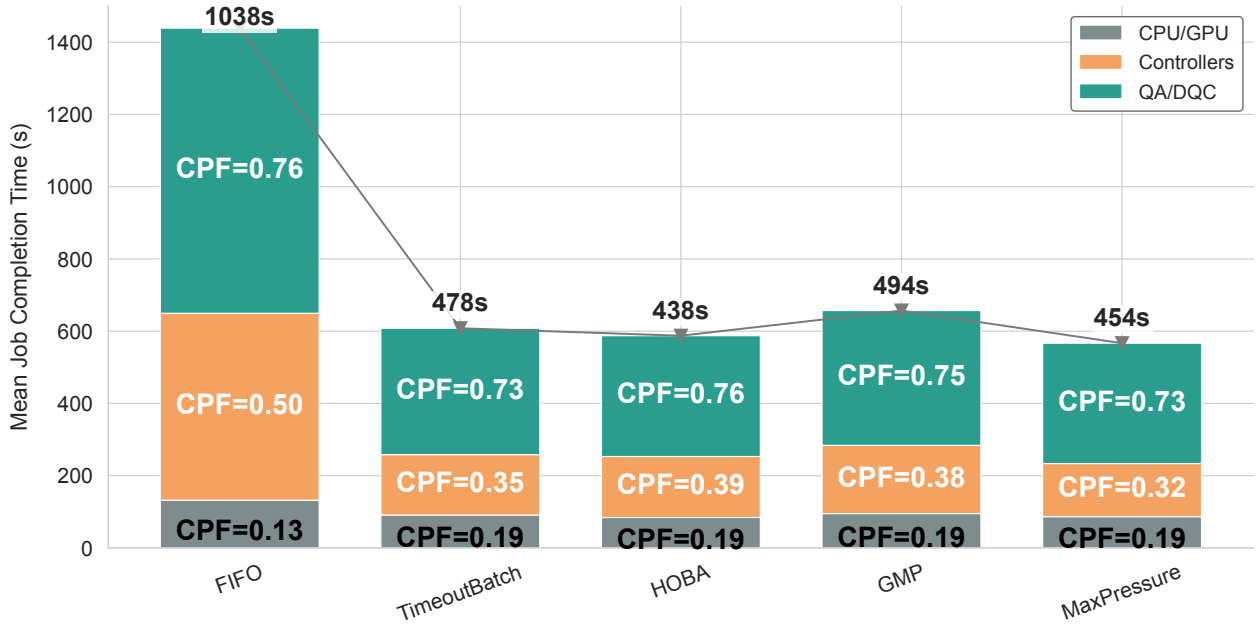


Figure S1: Original policy time composition. Labels above the bars give mean job-completion time. CPF is the critical path fraction diagnostic defined in the paper. Resource activity can overlap, so component sums are not elapsed job time. Policy definitions appear in the main paper.

The useful policy depends on the workload

Workload	Selected comparison	Runtime (s)
Bin packing	GMP / FIFO	0.615 / 0.732
Portfolio	HOBA / ABO or Max-Pressure	66.324 / about 75.9
PES scan	Max-Pressure / FIFO	1223.917 / 2403.988
Mixed bundle	Max-Pressure / HOBA / FIFO	2620.508 / 2685.251 / 3534.492

The mixed bundle retains an average DQC queue of 11.07-13.38 jobs across the reported policies. Better ordering reduces completion time while substantial demand remains queued. These workload-specific comparisons are separate from the main paper's 120-job stress study.

Which hardware upgrade helps?

Bin packing: 0.630 s baseline, 0.326 s with controller replication, and 0.225 s with balanced 10× scaling. Portfolio: 66.324 s baseline, 37.470 s with DQC scaling, and 26.857 s balanced. PES scan: 1418.809 s baseline, 1009.149 s with controller scaling, and 414.434 s balanced. Mixed bundle: 2685.251 s baseline, 1778.419 s with DQC scaling, and 1226.074 s balanced.

QA-only and faster-link-only upgrades leave portfolio at 66.324 s. The useful upgrade follows the active resource path. Balanced scaling helps all four cases, with gains of 2.19-3.42×.

Data: `topology_and_scheduling/policy_time_components.csv`, `topology_and_scheduling/policy_comparisons.csv`, `topology_and_scheduling/topology_comparisons.csv`. All paths are relative to the supplementary dataset [1].

2. Deployment overhead and hardware data

Cloud versus integrated execution

This study applies measured D-Wave wall/access factors to simulated QA service. A ratio is cloud-proxy makespan divided by integrated-HCU makespan. A delta is the added elapsed time. The D-Wave target determines the QA overhead model. DQC-only jobs therefore remain unchanged in this comparison.

QA target / job type	1× **ratio / delta s**	10× **ratio / delta s**	100× **ratio / delta s**
Advantage 1 / QA-only	2.540 / 3.78	2.758 / 37.82	2.784 / 378.20
Advantage 1 / QA+DQC	1.007 / 2.40	1.000 / 24.00	1.000 / 240.00
Advantage 2 / QA-only	4.045 / 7.48	4.476 / 74.80	4.527 / 748.00
Advantage 2 / QA+DQC	1.014 / 4.75	1.001 / 47.46	1.000 / 474.60
Either / DQC-only	1.000 / 0.00	1.000 / 0.00	1.000 / 0.00

QA-only workloads are sensitive to this overhead. In coupled jobs, a small ratio can still accompany a substantial absolute delta because DQC work dominates the total. Ratios rounded to 1.000 do not imply an exactly zero difference. Physical device occupancy and provider turnaround therefore remain separate model inputs.

How to read the grounding evidence

IBM measurements record gate-based circuits, shots, qubits, logical and transpiled depth, gate counts, and IBM-reported quantum seconds. A circuit measurement and a provider-billed Runtime job are different counting units because one Runtime job can contain several circuits. Backend-specific coefficients and fit metrics are provided in the archive.

The D-Wave runtime grids cover Advantage 1/Pegasus and Advantage 2/Zephyr, 64-4096 variables, degrees 2, 4, and 6, 100/1000/5000 reads, and 20/100 μs anneals. The retained training corpus contains 249 Advantage 1 and 263 Advantage 2 observations, including 39 and 28 unsuccessful embeddings. Embedding maps a problem onto the device's physical connectivity. Its failures supply feasibility evidence separately from successful timing measurements.

Workload-shaped validation adds application inputs. Further D-Wave studies vary portfolio construction, chain strength, and spin-reversal gauges. The archive separates campaign coverage, workload measurements, embedding statistics, timing components, model targets, coefficients, and fit metrics, so readers can inspect the evidence needed for a particular profile without treating all measurements as one fit.

Data: `cloud_execution/execution_mode_comparison.csv`, `hardware_measurements/ibm_measurement_campaigns.csv`, `hardware_measurements/dwave_campaign_parameters.csv`, `hardware_measurements/runtime_model_targets.csv`, `hardware_measurements/runtime_model_fit_metrics.csv`, `hardware_measurements/runtime_model_coefficients.csv`. All paths are relative to the supplementary dataset [1].

3. Workload growth and completion

The main paper separates data, circuit, shot, depth, and combined growth axes. A complementary study groups configurations by execution type and compares current workloads with larger workloads on an enlarged HCU. The entries below are medians over completed configurations.

Execution type	**Current HCU,** 1×	**Larger HCU,** 10× **workload**	**Larger HCU,** 100× **workload**
QA-only	2.57 s	22.74 s	225.6 s
DQC-only	470.4 s	64,841 s	52.76 million s
QA+DQC	385.6 s	74,728 s	64.26 million s

Each scale completes 108/108 QA-only, 162/162 DQC-only, and 324/324 coupled configurations. QA-only runtime grows roughly with the workload factor. DQC and coupled runtime grows much faster when circuit count, shots, and depth increase together. These factors multiply service demand while controllers and other dependent stages continue to contend.

Payload growth can move a different bottleneck

At 100× data-only growth, PES scan has a 20 GB maximum payload and remains CPU-limited. Its runtime falls from 5544 s on the fixed machine to 848.4 s on the larger machine. In contrast, shot-only VQE remains DQC-limited, taking 46,654 s on the fixed machine and 21,067 s on the larger machine. The growth axis determines which resource merits attention.

Completion status must accompany elapsed time

On the fixed HCU, the broad 10× study completes 162/594 configurations, leaves 48 partial configurations, and has 384 configurations with no completed jobs. At 100×, it completes 0/594 configurations. A configuration can contain many submitted jobs.

Extended-runtime follow-up, 10×	**Completed configurations**	**Median time of completed cases**
QA-only	108 / 108	22.74 s
DQC-only	18 / 162	59,808 s
QA+DQC	36 / 324	99,372 s

The coupled group also retains 48 partial configurations. At 100×, the follow-up still reports no completed configurations. Longer execution alone does not establish feasibility for the configurations that remain unfinished. Their elapsed times are not used as completed-run speedups.

Representative 100× all-field cases show the job-level distinction: cloud scheduling completes 28/30 jobs, pipeline 15/30, transmission switching 13/30, unit commitment 17/30, and VQE 0/30 under the fixed-HCU settings. The four partial cases report DQC utilization of 1.000.

Data: `workload_scaling/hcu_scaling.csv`, `workload_scaling/workload_growth.csv`, `workload_scaling/fixed_hcu_completion_counts.csv`, `workload_scaling/extended_runtime_followup.csv`, `workload_scaling/fixed_hcu_workloads.csv`. All paths are relative to the supplementary dataset [1].

4. Controller and link sensitivity

The targeted studies vary communication links and latency jitter, cross-fabric control paths, controller/device links, and resource replicas. They use VQE (variational quantum eigensolver) and PES (potential-energy-surface) scan workloads to examine regimes that an aggregate workload mean can hide.

Current workloads on the fixed HCU

Sweep	PES scan completed / tested	VQE completed / tested
Communication links / jitter	240 / 240	240 / 240
Cross-fabric control	90 / 115	90 / 115
Controller / device links / replicas	240 / 240	240 / 240

Across these groups, median runtimes range from 3440.0 to 3665.3 s for PES scan and from 3320.1 to 3343.3 s for VQE. The 25 unfinished configurations in each cross-fabric sweep remain distinct from completed timing comparisons.

Larger workloads need sufficient machine capacity

At 10× workload scale on the fixed HCU, PES scan completes no jobs. VQE has a median of four completed jobs out of fifty in the grouped configurations. At 100×, neither workload reaches job completion in these targeted studies.

Workload scale on larger HCU	PES scan median runtime	VQE median runtime
10×	about 38,750 s	about 260,935 s
100×	about 5.44 million s	about 208.88 million s

Both enlarged-HCU studies complete 1172/1222 tested configurations. At 100×, PES scan has median DQC utilization of about 0.926-0.927, while VQE has about 0.544. These values describe the reported groups, with the incomplete configurations retained separately.

Architectural decision

First establish that the base machine can carry the selected workload. Then compare controller placement, routing, and link parameters among completed matched configurations. If the workload is already unable to complete under the machine's capacities, a fine-grained latency sweep cannot by itself establish a useful upgrade.

A topology search applies the same principle over combinations of resources. It adds explicit devices, controllers, compute pools, memory, or links and records whether the bottleneck is reduced or moves elsewhere. The workload definitions and scheduling policy must stay fixed when attributing a difference to the topology.

Data: `sensitivity_studies/controller_link_sensitivity.csv`, `system_configuration/controller_variants.csv`, `system_configuration/link_profiles.csv`. All paths are relative to the supplementary dataset [1].

5. QA sampling and load shedding

The QA success model uses 453 saved D-Wave SampleSet runs containing 926,900 samples. A hit lies within 1% of the best observed energy in the same problem size/degree group. Repeated copies of a returned sample are weighted by their occurrence counts.

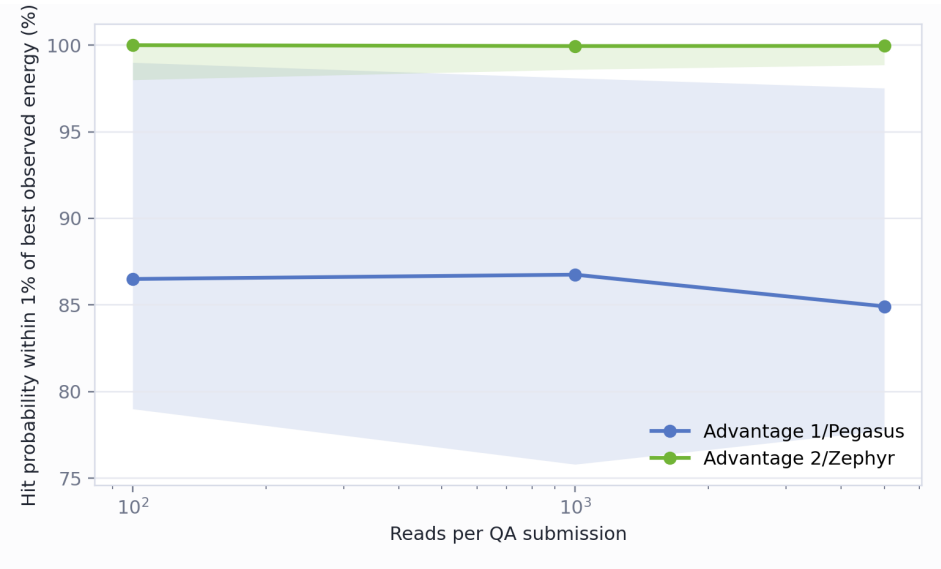


Figure S2: Original QA success plot at a 20 μs anneal. Bands show the interquartile range across problem groups. Advantage 2 is near saturation at this target, while Advantage 1 retains appreciable fallback probability.

When does more QA work pay off?

For sample x with energy $E(x)$, group best energy $E^\star$, and tolerance $\epsilon = 0.01$, a hit satisfies

$$\frac{E(x) - E^\star}{\max(1, |E^\star|)} \leq \epsilon. \tag{S3}$$

Let p be the occurrence-weighted hit probability. Under the independent-attempt approximation, A attempts have success probability $P_{\text{success}}(A)$ and fallback probability $F(A)$:

$$P_{\text{success}}(A) = 1 - (1-p)^A, \qquad F(A) = (1-p)^A. \tag{S4}$$

Increasing the cap from A_0 to A_1 pays off in expected service when the DQC fallback time T_{DQC} satisfies

$$T_{\text{DQC}} > \frac{T_{\text{QA}}(A_1) - T_{\text{QA}}(A_0)}{F(A_0) - F(A_1)}, \qquad F(A_0) > F(A_1), \tag{S5}$$

where $T_{\text{QA}}(A)$ is the QA time at cap A. Backlog-aware load shedding also considers the DQC waiting time.

Mode on current HCU	Relative QA cap	Median runtime (s)
Load shedding	1× / 8× / 16×	156.4 / 7.43 / 5.30
Quality fixed	1× / 4× / 16×	151.8 / 167.5 / 19.84

Load-shedding throughput rises from 0.192 to 4.04 and 6.17 jobs/s across the three caps. Extra QA effort can remove enough DQC fallback work to help, but a 4× quality-fixed cap first makes this study slower. At larger workload scales, the reported load-shedding runtime stays near 66,743 s at 10× and 56.99 million s at 100× across caps. The success curve matters when it changes the limiting path.

Data: `hardware_measurements/qa_success_dataset.csv`, `hardware_measurements/qa_success_by_reads.csv`, `sensitivity_studies/qa_amplification_sensitivity.csv`. All paths are relative to the supplementary dataset [1].